

Last time $D \subseteq \mathbb{R}$ domain
 $S_D := \{f: D \rightarrow \mathbb{R}\}$

$$+ : (f+g)(x) = f(x) + g(x)$$

$$\cdot : (f \cdot g)(x) = f(x) \cdot g(x)$$

We looked at P_1 - P_9 .

P₁₀. For all $f \in S_D$, $\exists P \subseteq S_D$ s.t. exactly one of these is true.

$$\begin{cases} f=0 \text{ (} 0(x)=0 \text{ } \forall x) \\ f \in P \\ -f \in P \end{cases}$$

Prop P_{10} - P_{12} fails for $(S_D, +, \cdot)$

Pf. let $a, b \in D$ be two distinct points on D .

let $f: D \rightarrow \mathbb{R}$

$$x \mapsto \begin{cases} 1 & \text{if } x=a \\ 0 & \text{if } x \neq a. \end{cases}$$

$g: D \rightarrow \mathbb{R}$

$$x \mapsto \begin{cases} 1 & \text{if } x=b \\ 0 & \text{if } x \neq b. \end{cases}$$

Then either f or $-f \in P$
 g or $-g \in P$

But $f \cdot g, (-f) \cdot g$
 $f \cdot (-g), (-f) \cdot (-g)$ are all 0.

By P12, if $\pm f, \pm g \in P$
then $(\pm f)(\pm g) \in P$
but they are all 0 regardless of signs.

$\Rightarrow$ P12 fails for $(S_0, +, \cdot)$.

$$S := \{f: D \longrightarrow D\}$$

$$(f+g)(x) = f(x) + g(x)$$

$$(f \circ g)(x) = f(g(x)).$$

Identity if $f \in S_D$,

$$(I \circ f)(x) = I(f(x)) = f(x).$$

$$(f \circ I)(x) = f(I(x)) = f(x).$$

So I is the multiplicative identity.

Mul. inv. If $f \in S_D$

We want $f^{-1}(x)$ to be the mult. inv.

But in order for $f^{-1}(x)$ to even exist,

$f: D \rightarrow D$ has to be one-to-one

(eg: $f: D \rightarrow D$
 $x \mapsto 1$).

Correspondence

$f^{-1}(f(x)) = f(f^{-1}(x)) = I(x) = x$ only makes sense if f is 1 to 1.

Distributive doesn't hold

$$f \circ (g+h)$$

$$\text{let } f(x) = \sin x$$

$$g(x) = h(x) = x$$

$$\text{LHS: } (f \circ (g+h))(x)$$

$$= f((g+h)(x))$$

$$= f(g(x) + h(x))$$

$$= f(2x)$$

$$= \sin 2x$$

$$\text{RHS: } f \circ g + f \circ h(x)$$

$$= (f \circ g)(x) + (f \circ h)(x)$$

$$= f(g(x)) + f(h(x))$$

$$= f(x) + f(x)$$

$$= 2 \sin x$$

$$\therefore f \circ (g+h) \neq (f \circ g + f \circ h)(x).$$

Ex ① let $I(x) = x$ identity

$$f(x) = \sin x$$

How to express $\frac{x + x^2 + x \sin^2 x + \sin(\sin x y)}{x \sin x}$ using just f, I ?

Solution. $\left(\frac{I + I^2 + I \cdot f \cdot f + f \circ f \circ (I \cdot I)}{I \cdot f} \right) (x)$

② $I(x) = x$

$$S(x) = \begin{cases} 1, & x \geq 0, \\ -1, & x < 0. \end{cases}$$

Express $|x|$ using I and S .

Solution. $S \cdot I(x)$
 $-S \cdot I(x)$

IV. Graphs.

Notation

$\mathbb{R}$

What it is

Real #

$|a|$

abs. value

$|a-b|$

abs. value

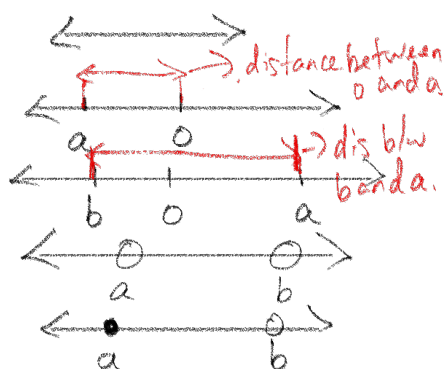
(a, b)

Open interval

$[a, b)$

closed / open

Picture



$|PQ|$
 $\|PQ\|$

distance b/w, P, Q

